

Report on Faraday's Law of Electromagnetic Induction

Self-Visit to an Electrical Repair Shop

Report Prepared by: Mansi Subhash Zalte

Date of Visit: 5th Oct 2025

Location: Ozar (Mig)

Introduction

Faraday's Law of Electromagnetic Induction states that *whenever the magnetic flux linked with a coil changes, an electromotive force (EMF) is induced in the coil*. This principle is the basis of transformers, motors, generators, and many household appliances. To connect theory with real-world practice, I visited a nearby electrical repair shop Samarth Electricals Ozar (Mig) on 5th Oct 2025 and observed how this law is applied in devices we use every day. . The shopkeeper, Mr. Jayesh Kolapkar , kindly allowed me to examine the appliances and explain how they work.

Purpose of Visit

- To see practical applications of Faraday's Law in electrical appliances.
- To understand how technicians deal with problems related to electromagnetic induction.
- To relate classroom knowledge with field observations.

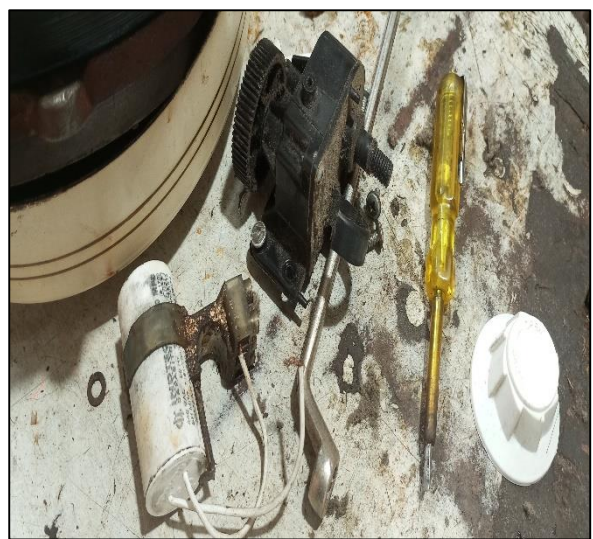
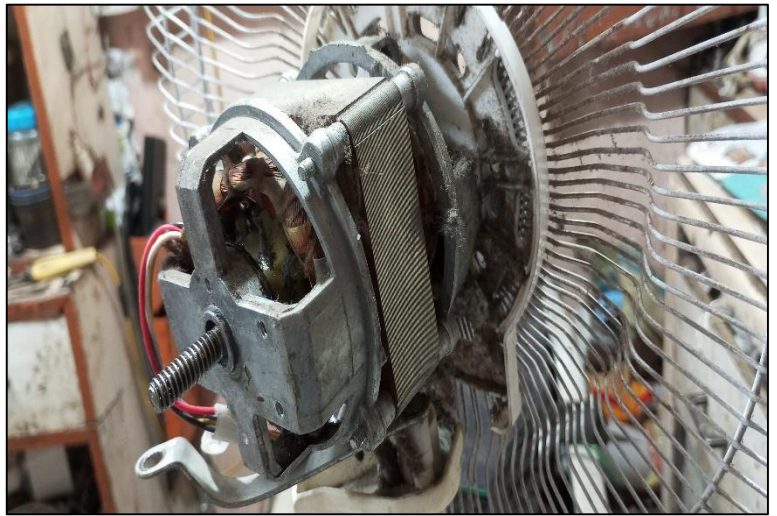
Observations

During my self-visit, I noticed the following:

- 1. Electric Motors (Fans and Pumps)**
 - I saw the technician repairing a ceiling fan motor. He showed me the copper windings that had burnt out and explained that when current passes through these coils, electromagnetic induction helps in producing rotation.
 - He also mentioned that faulty windings reduce efficiency, so rewinding is a common repair.
- 2. Transformers**
 - The shop had stabilizers and chargers under repair. The technician explained that these contain small transformers which work on the principle of electromagnetic induction. A varying current in the primary coil induces voltage in the secondary coil.
- 3. Induction Stove**
 - An induction cooktop was being checked. The technician explained that it generates a changing magnetic field, which induces eddy currents in the cooking vessel. These currents produce heat for cooking.

4. Portable Generator

- I observed a small generator under servicing. The owner explained that when the generator's coil rotates in a magnetic field, electricity is induced, which is exactly Faraday's Law in action.



Learning and Discussion

From my visit, I realized that Faraday's Law is not just theoretical—it is at the heart of nearly every electrical device. Most repair work in the shop was related to coils, windings, and magnetic components, which directly connect to the principle of induction. Seeing real devices strengthened my understanding of how changing magnetic fields create useful electrical energy.



Conclusion

The self-visit gave me valuable practical insights into Faraday's Law of Electromagnetic Induction. From motors to transformers and generators, I saw clear evidence of this principle in action. The experience helped me connect my textbook knowledge with real-life applications.